

LAB SETUP REPORT

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1. Objective

To set up a cybersecurity lab environment using Kali Linux, Metasploitable 2, and Wireshark for ethical hacking, vulnerability assessment, and network traffic analysis.

2. Lab Configuration

Component	Details
Virtualization Software	Oracle VirtualBox
Attacker Machine	Kali Linux Purple
Target Machine	Metasploitable 2
Kali IP	192.168.1.3
Metasploitable IP	192.168.1.4
Packet Analyzer	Wireshark

3. Screenshots and Analysis



Figure 1: Kali Linux Operating System Successfully Installed

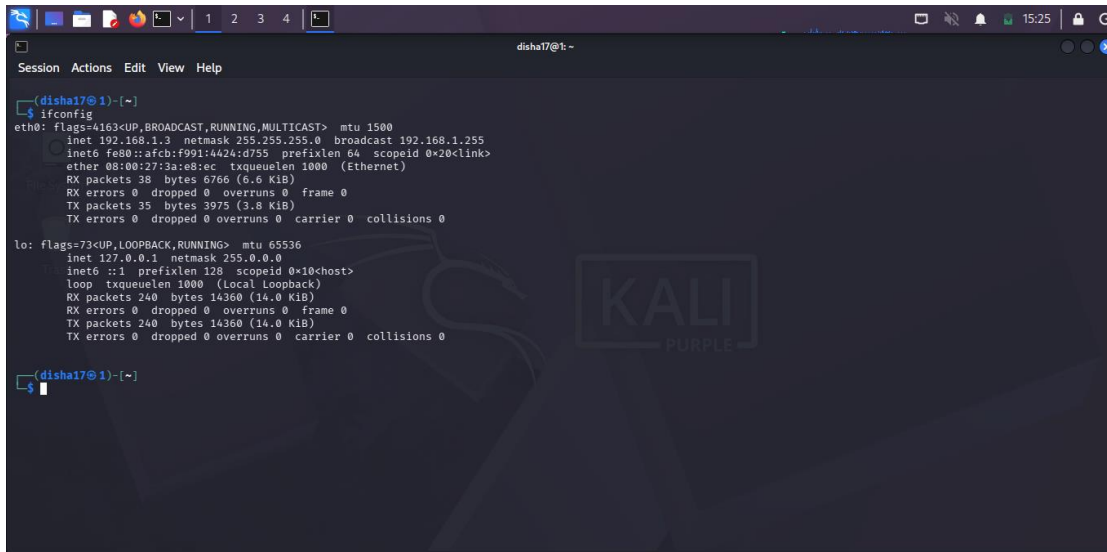


Figure 2: Network Configuration of Kali Linux Showing IP Address 192.168.1.3

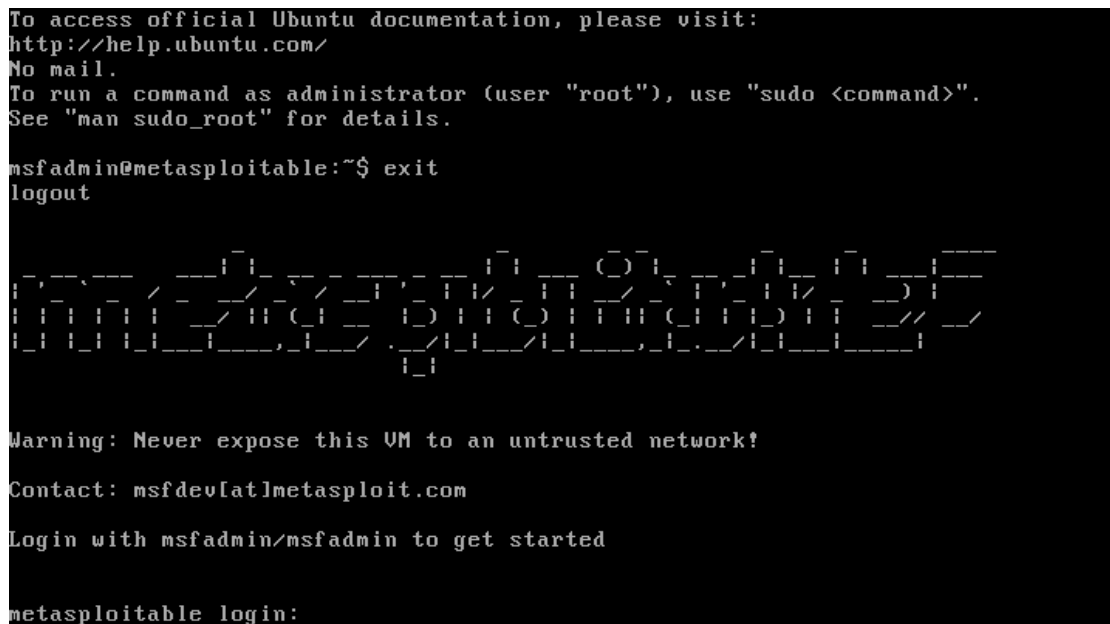


Figure 3: Metasploitable 2 Virtual Machine Boot Screen

```

http://help.ubuntu.com/
No mail.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

msfadmin@metasploitable:~$ exit
logout

      _ _ _ _ _
     / _ _ _ _ \
    / _ _ _ _ \
   / _ _ _ _ \
  / _ _ _ _ \
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Warning: Never expose this VM to an untrusted network!

Contact: msfdev[at]metasploit.com

Login with msfadmin/msfadmin to get started

metasploitable login: msfadmin
Password: _

```

Figure 4: Successful Login Using Default Credentials (msfadmin/msfadmin)

```

No mail.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:16:09:eb
          inet addr:192.168.1.4  Bcast:192.168.1.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe16:9eb/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:49 errors:0 dropped:0 overruns:0 frame:0
          TX packets:70 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:6818 (6.6 KB)  TX bytes:7828 (7.6 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:108 errors:0 dropped:0 overruns:0 frame:0
          TX packets:108 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:27569 (26.9 KB)  TX bytes:27569 (26.9 KB)

msfadmin@metasploitable:~$ _

```

Figure 5: Network Configuration of Metasploitable 2 Showing IP Address 192.168.1.4

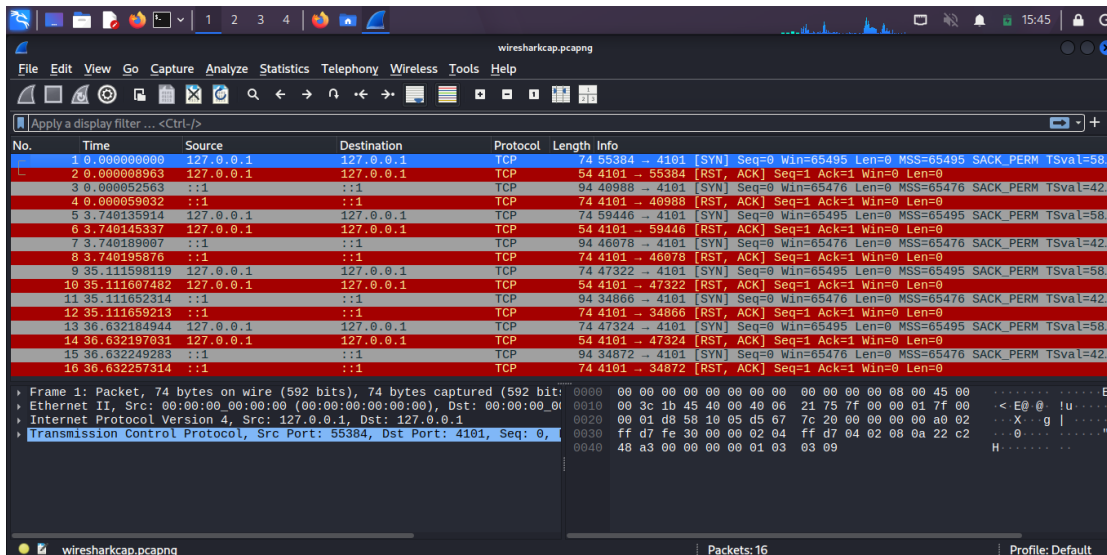


Figure 6: Network Traffic Captured and Analyzed Using Wireshark

4. Observations

- Kali Linux was successfully installed and configured.
- Kali Linux IP Address: 192.168.1.3
- Metasploitable 2 was successfully deployed.
- Metasploitable 2 IP Address: 192.168.1.4
- Wireshark successfully captured TCP traffic.
- The lab environment is ready for penetration testing exercises.

5. Commands Used

```
ifconfig
sudo wireshark
ping 192.168.1.4
nmap 192.168.1.4
```

6. Conclusion

The cybersecurity lab environment was successfully established using Kali Linux, Metasploitable 2, and Wireshark. Network connectivity and packet capture were verified, providing a controlled environment for ethical hacking and security testing.